

APOLLO INSTITUTE OF COMPUTER APPLICATION
BACHELOR OF COMPUTER APPLICATION
SEMESTER 03
Design Thinking and Innovation (BC03001071)
Lecture Notes
Unit 3: Defining Problems (5 Hours)

LECTURE 1: KNOW YOUR PROBLEM

1. Introduction

In Design Thinking, after **Empathizing with users and conducting user research**, the next step is to **define the problem**.

The purpose of this stage is to understand the **real problem** faced by the user.

Many times, we notice a problem and immediately think about a solution. Design Thinking says:

First understand the problem properly, then think about the solution.

Example

Suppose students frequently miss college notices.

A quick solution might be:

"Let's create a mobile app."

But this may not be the actual problem.

The actual problem could be:

"Students do not receive important information at the right time."

Once we understand the real problem, we can think of many possible solutions.

2. What is a Problem?

A **problem** is a difficulty, challenge, or unmet need experienced by a user.

Example

- Students wait for a long time at the college office.
- Users find a website difficult to navigate.
- Customers have difficulty making online payments.
- Students cannot easily find study materials.

3. What Does "Know Your Problem" Mean?

Knowing your problem means understanding:

- **Who** is facing the problem?
- **What** problem are they facing?
- **When** does the problem occur?
- **Where** does it occur?
- **Why** does the problem occur?
- **What** does the user actually need?

This helps us move from a **general problem** to a **specific problem**.

4. Problem vs Solution

It is important to distinguish between a **problem** and a **solution**.

Problem

Students find it difficult to receive college notices on time.

Solution

Create a mobile application for students.

The first statement explains the **difficulty**.

The second statement gives a **possible solution**.

In Design Thinking, we should define the problem before selecting a solution.

5. Problem Statement

A **problem statement** is a short and clear description of the problem faced by the user.

Example

"College students find it difficult to receive important notices on time."

A good problem statement should be:

- Clear
 - Specific
 - User-centered
 - Based on research
 - Focused on the actual problem
 - Free from a specific solution
-

6. Steps to Define the Problem

Step 1: Identify the User

Who is experiencing the problem?

Step 2: Understand the Difficulty

What difficulty does the user face?

Step 3: Study the Situation

When and where does the problem happen?

Step 4: Find the Root Cause

Why is the problem happening?

Step 5: Identify the Need

What does the user actually need?

Step 6: Write the Problem Statement

Express the problem in a short and clear sentence.

7. Example

Situation: Students wait in a long queue to submit forms.

Observation: Students spend a lot of time at the office.

Problem: Students waste time waiting to complete simple administrative tasks.

Need: Students need a faster and more convenient process.

Problem Statement:

"Students need a convenient way to complete routine administrative tasks without spending excessive time waiting in queues."

Key Point

Do not solve the problem before understanding it.

LECTURE 2: INSIGHTS

1. Introduction

During user research, we collect a lot of information through:

- Interviews
- Surveys
- Observation
- Questionnaires
- Conversations
- Feedback

However, simply collecting information is not enough.

We need to understand **what that information means**.

This deeper understanding is called an **insight**.

2. What is an Insight?

An **insight** is a meaningful understanding of a user's:

- Needs
- Feelings
- Behavior
- Motivation
- Difficulties

An insight helps us understand **why the user behaves in a particular way**.

Easy Definition

An insight is a deeper understanding discovered from user research.

3. Observation vs Insight

This is an important difference.

Observation

Students frequently check their phones.

This tells us **what they do**.

Insight

Students frequently check their phones because they are worried about missing important updates.

This tells us **why they do it**.

Therefore:

Observation = What?
Insight = Why?

4. How Are Insights Generated?

Insights can be identified by:

1. Finding Patterns

Look for behavior repeated by many users.

2. Identifying User Needs

Understand what users actually require.

3. Understanding Emotions

Identify feelings such as frustration, fear, confusion, or satisfaction.

4. Finding Hidden Problems

Sometimes users may not directly tell us their real problem.

5. Asking "Why?"

Repeatedly asking why can help discover the root cause.

5. Example

Suppose we observe:

Students frequently ask teachers about assignment deadlines.

Observation

Students repeatedly ask about deadlines.

Possible insight

Students may not have a reliable way to remember or track deadlines.

User need

Students need an easy way to receive and remember deadline information.

6. Characteristics of a Good Insight

A good insight should be:

- **User-centered**
- **Based on evidence**
- **Meaningful**
- **Specific**
- **Relevant**
- **Actionable**

An insight should help the design team decide **what kind of problem needs to be addressed**.

7. Example from Daily Life

Observation:

People leave shopping websites without completing purchases.

Possible insight:

Users may leave because they are uncertain about delivery charges or return policies.

Need:

Users need clear information before making a purchase.

This insight can help designers improve the shopping experience.

Key Point

Insights help us move from "what users do" to "why users do it."

LECTURE 3: POINT-OF-VIEW (POV)

1. What is POV?

POV stands for **Point-of-View**.

A **Point-of-View statement** defines the problem from the perspective of a specific user.

It combines:

User + Need + Insight

POV helps the design team focus on the **right user and the right need**.

2. Basic Format of POV

A simple format is:

[User] needs [Need] because [Insight].

3. Example

User:

College students

Need:

Easy access to important notices

Insight:

Students often miss information displayed on physical notice boards.

POV:

College students need easy access to important notices because they often miss information displayed on physical notice boards.

4. Components of POV**1. User**

Who are we designing for?

Example:

College students

2. Need

What does the user need?

Example:

Easy access to important information

3. Insight

Why does the user need it?

Example:

They often miss important information from physical notice boards.

5. Why is POV Important?

POV helps us:

- Focus on the user's needs.
- Clearly define the problem.
- Use research findings.
- Avoid assumptions.
- Guide the ideation process.
- Develop meaningful solutions.

6. Bad POV vs Good POV**Bad POV**

We need to create a mobile application for students.

Why is it bad?

Because it focuses on a **solution**.

Good POV

Students need an easy way to receive important notices because they often miss information displayed on physical notice boards.

This focuses on:

User → Need → Insight

7. Another Example

Problem:

Elderly users find online banking difficult.

POV:

Elderly users need a simple way to complete online banking tasks because complex interfaces can make them confused and uncomfortable.

Key Point

POV tells us who has the problem, what they need, and why they need it.

LECTURE 4: HMW – HOW MIGHT WE QUESTIONS

1. Introduction

Once the problem has been clearly defined through a POV statement, the next step is to create a question that encourages **creative thinking**.

This is done using **HMW questions**.

2. What is HMW?

HMW stands for:

How Might We

An HMW question converts a problem into an **opportunity for generating ideas**.

Definition

An HMW question is an open-ended question that encourages the design team to think of multiple possible solutions to a user problem.

3. Meaning of Each Word

How

Encourages us to think about possible methods.

Might

Shows that there can be many possible answers.

We

Encourages teamwork and collaboration.

4. Basic Format

How might we + help the user + solve the problem?

5. Example

Problem:

Students miss important college notices.

HMW:

How might we help students receive important college notices on time?

This question can lead to many ideas:

- Mobile app
- SMS alerts
- Email notifications
- WhatsApp notifications
- Digital notice board

6. Another Example

Problem:

Students spend too much time waiting at the college office.

HMW:

How might we reduce students' waiting time at the college office?

Possible ideas:

- Online appointment system
- Digital token system
- Online form submission

- Queue management system

7. Characteristics of a Good HMW Question

A good HMW question should be:

Open-ended

It should allow multiple answers.

User-centered

It should focus on the user's need.

Clear

It should be easy to understand.

Specific

It should address a particular problem.

Solution-neutral

It should not force one particular solution.

8. Bad and Good HMW

Bad

How might we create a mobile app for students?

This already decides the solution.

Good

How might we help students access college information more easily?

This allows different solutions.

Key Point

HMW converts a problem into an opportunity for creative ideas.

LECTURE 5: CONNECTING PROBLEM, INSIGHT, POV AND HMW

This lecture is important because students should understand **how all four concepts work together.**

1. Complete Design Thinking Flow

The process can be remembered as:

Research → Problem → Insight → POV → HMW → Ideas

Each stage has a different purpose.

2. Complete Example

Step 1: Research

We interview college students and observe how they receive college information.

Step 2: Problem

Students often miss important college notices.

Step 3: Observation

Students frequently ask classmates about important announcements.

Step 4: Insight

Students depend on classmates because they do not have a reliable and convenient source for receiving updates.

Step 5: POV

College students need a reliable and easy way to receive important information because they currently depend on others for updates.

Step 6: HMW

How might we help college students receive reliable information quickly and easily?

Step 7: Ideas

Now the team can brainstorm:

- College mobile application
- SMS notification system
- Email alerts
- WhatsApp notification system
- Digital notice board
- Student portal

Relationship Between All Four Concepts

Problem

What is wrong?



Insight

Why is it happening?



POV

Who has the problem + What do they need + Why?



HMW

How might we help them?



Ideas

What possible solutions can we create?

Easy Formula for Students

PROBLEM

What?

INSIGHT

Why?

POV

Who + Need + Why?

HMW

How might we?

IDEATION

What ideas can solve it?

Summary

Lecture	Topic	Main Focus
1	Know Your Problem	Identify and clearly define the actual user problem
2	Insights	Understand the deeper meaning behind user behavior
3	Point-of-View	Define the problem from the user's perspective
4	HMW Questions	Convert the problem into an open-ended design question
5	Connecting All Concepts	Apply Problem → Insight → POV → HMW → Ideas